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# An energy-aware drone trajectory planning scheme for terrestrial sensors localization

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ABSTRACT

Employing GPS-equipped drones to act as mobile anchors is a popular solution for terrestrial sensors positioning in a generic environment. Researchers have proposed several approaches, usually to reduce estimated locations error and increase localization coverage, though no efficient solution has been presented for energy conservation of the drone. Drones, such as the commercial quadcopters, have limited power supply and cannot fly long. Any localization algorithm should consider the energy constraints beside the performance indicators. Furthermore, there is no suggested strategy to mitigate the error of Received Signal Strength (RSS)-based distance measurements in the existing solutions. In this paper, we propose a novel scheme to plan the drone trajectory, called the "Weighted Energy-aware Trajectory with Adaptive Radius (*WETAR*)". The proposed scheme employs Linear Programming (LP) for trajectory planning in the presence of the sensors with given estimative regions which are acquired in a range-free pre-localization phase. It also specifies candidate waypoints, which are the projection of aerial anchor points on the ground, and assign weights to them based on two criteria: quality of the beacons that the sensors would receive and their coverage ratio. We assume that sensors utilize a range-based localization algorithm on the basis of RSS measurements. Simulation results show that the *WETAR* as an aerial anchor guiding mechanism, guides the drone effectively and reduces localization time, saves the drone energy, and improves the location error as well as the localization coverage.

## 1. Introduction

In WSNs each node is equipped with small sensors to monitor environment condition. These WSNs are the basis of the emerging technology of Internet of Things (IoT) and specifically fit very well in surveillance applications [1]. They are integrated into the IoT to serve in applications like automation, environment or health monitoring and smart grids. Localization is one of the major issues in WSNs because the acquired information from each node is not useful until its location is known [2]. The most plausible solution to the localization problem is deploying sensor nodes that are aware of their own location via GPS or installing them at fixed positions. These sensors that are called static anchors help unknown sensors to discover their location based on the information provided by the anchors in the broadcasted beacons [3]. The anchor-based localization schemes are categorized into two types; namely range-based and range-free [4]. In range-based localization techniques, range measurements are done based on the Time of Arrival (ToA) [5–8], Time Difference of Arrival (TDoA) [9,10], Angle of Arrival or the Received Signal Strength (RSS) [11–13]. Unknown sensors then use this range information in the localization process. Range-based techniques require additional hardware to estimate the distance or angle between sensors. However, the range-free ones use graph information and the connections between sensors and anchors without the ranging information thus are less equipment-demanding [3,14–16].

Among range-based schemes, RSS-based techniques are straightforward to implement, unlike the T(D)oA-based and AoA-based techniques that need highly accurate clocks and antenna arrays on board [17]. The simplicity of RSS is attractive in low-cost and resource-constrained WSNs.

In WSNs that inaccurate measurements yield poor performance, deploying a higher number of stationary anchors will lead to better performances, but it is not a cost efficient solution. A consensus has been reached in the previous researches [18–27] on the introduction of mobile anchor-assisted positioning as a cost-effective solution to the above problem. An Unmanned Arial Vehicle (UAV) which is equipped with a GPS can be utilized as the mobile anchor. Recently, aerial mobile anchors, also known as drones, have had widespread use in localization of WSNs employed for different applications [28]. For example, drones can be utilized to localize sensors deployed in rescue operations after a

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disaster like earthquake, bushfire or avalanche happens. Sensors can be randomly deployed in the disaster area and a drone can hover above the harsh or inaccessible environment to help sensors estimate their locations.

During UAV operation, the drone flies over the deployment area and broadcasts beacons that include its coordinates and transmission power. The covered terrestrial nodes calculate their distances from the aerial anchor points using an RSS-based technique, which are referred to as slant distances. The projection of aerial anchor points and slant distances on the terrestrial monitoring environment are called waypoints and ground distances, respectively. After collecting at least three non-collinear beacons, each sensor with unknown location uses the measured ground distances in multilateration/trilateration approach to estimate its position.

Two major issues in drone-based localization are localization technique and trajectory planning. In this paper, we focus on drone trajectory planning for an acceptable level of positioning performance since the flight time of a drone is quite limited. Drone energy cannot be infinite in many scenarios e.g. disaster management. This makes energy consumption a critical issue in trajectory planning. Another important issue which should be taken in to account in path determination is how to improve distance measurement accuracy in a drone-based communication system while using an RSS-based technique.

In this paper, a novel path planning scheme is proposed for an energy-constrained drone to construct the so called Weighted Energy-aware Trajectory with Adaptive Radius (*WETAR*). The coverage ratio of sensors and the distance between the receiver sensors and the drone projection on the ground at each waypoint are considered as determinant factors in weight definition of waypoints; which manifest the quality of beacon that can be provided at each location.

The contributions of this article are summarized below:

- We consider and formulate the energy limitation of the drone in its guiding approach for range-based localization.
- We provide an expression for RSS-based distance estimation error based on the real distance.
- We assign weights to waypoints. The number of covered sensors at each waypoint and the quality of beacons that it can provide for the sensors are defined as metrics to calculate such weights.
- Then, we find an optimal path and optimal transmission powers for each selected waypoint by linear programming.

The rest of the paper is organized as follows: Section 2 reviews the existing literature on mobile anchor-based localization. Section 3 introduces the new guiding scheme for the drone. In Section 4 we present and discuss the experimental evaluations, and finally, the conclusion is given in Section 5.

## 2. Related work

As the survey in [29] implies, UAV-based localization technique is more reliable than terrestrial mobile-anchor based techniques due to two reasons; first, terrestrial anchors are more likely to be threatened by environmental or man-made hazards, and second, LOS (Line of Sight) probability is higher in air to ground channels which further improves positioning accuracy.

Applying an efficient location determination scheme, and guiding the aerial anchor (i.e, the drone) properly are critical challenges in localization of terrestrial objects in both static and mobile sensor networks. Path planning of UAVs for localization of static sensors is investigated in this paper since this branch of sensor networks guarantees a wide range of applications. The RSS feature is utilized in this study to measure the ground distance between the drone and its covered sensors at each waypoint. These estimated distances are then used for localization by multilateration technique.

Current works on path-selection methods for a mobile anchor have categorized them into random walk, static path and dynamic path.

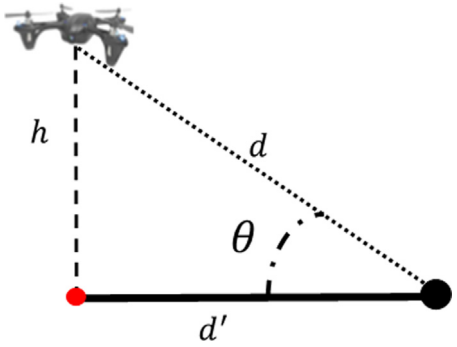



**Fig. 1.** Relationship between LOS probability and elevation angle.

In the static trajectory approach [18–23], the mobile anchor path is determined before it starts moving for localization. At the other side, a real-time dynamic path planning approach continuously (dynamically) does the planning during the localization process based on the actual distribution of unknown nodes [24,25]. This results in better location estimation precision especially in environments with non-uniform distribution of sensors. However, dynamic path planning increases energy consumption of sensors and anchor due to the more computations imposed.

In the rest of this section, we survey some related works in trajectory planning methods for terrestrial mobile anchors.

Three deterministic mobile anchor trajectories are presented in [19], namely SCAN, HILBERT and DOUBLE SCAN. Unlike the random walk, SCAN covers the network field uniformly. It divides the area to sub-regions and connects their centers using straight lines which together, form the mobile anchor path. It leads to full localization coverage if sensors are uniformly distributed. However, collinearity of beacon points in this simple trajectory affects the localization performance. SCAN is transformed into DOUBLE SCAN to mitigate collinearity at the expense of doubled trajectory length. HILBERT space-filling curves in [19] is another solution which ensures more path turns compared to SCAN. Therefore, it provides more non-collinear beacon points without significantly increasing the trajectory length. HILBERT trajectory may not be able to localize border nodes accurately, since they usually receive fewer beacons, and usually from one direction only, compared to the nodes located in the central region.

The idea of guiding the mobile anchor along the sides of an equilateral triangle and broadcasting beacons at its vertices was proposed in [12]. Then Han et al. in [20] proposed LMAT trajectory based on this configuration. LAMT aims to ensure that each sensor receives at least three non-collinear beacons and minimizes the location uncertainty of unknown nodes. Unknown sensors estimate their coordinates by trilateration where the distance between an unknown sensor and the mobile anchor is measured by RSSI.

Han et al. [21] considered the path planning problem as seeking LMAT-based beacon points [20] while traversing SCAN trajectory. The so called SLMAT method ensures that each sensor receives three non-collinear beacons from the vertices of a regular triangle. It also tries to conserves energy by reducing the number of corners along the planned path.

The proposed trajectory in [13], called Zcurve, follows Z-shaped curves. This Z-shape design is proposed to overcome collinearity problem with short jumps and to provide three consecutive non-collinear beacons for covered sensors in order to reduce the localization latency. In Zcurve the monitoring area is divided to square-like cells and the path is constructed by connecting the centers of those cells. Obstacle-handling is also considered in Z-shape trajectory.

Another static path planning algorithm which involves search-and-decide phases has been proposed in [22]. In this scheme number of the beacon points and their locations which form equilateral triangles to

offer complete coverage, are fixed. During the search-phase, the mobile anchor follows the SCAN trajectory and broadcasts its coordinate at some of the pre-determined beacon points to collect neighborhood information and identify regions which are occupied with at least one sensor. In the decision-phase, mobile anchor only broadcast beacons at a subset of the beacon points which are located in occupied regions. Thus, it elevates localization efficiency by avoiding useless beacon broadcasting in empty regions.

OPTEC algorithm, which has been proposed in [23] by Kouroshnezhad et al. is a recent static anchor guiding approach that utilizes initial region estimations of sensors for an optimal path planning. This study has additionally considered mobile anchor energy constraints during the planning phase. OPTEC also considers LMAT [20] beacon points as candidate coordinates and prioritizes them based on their contribution to location uncertainty reduction of sensors. It utilizes Integer Linear Programming to determine the optimal beacon points based on some defined priorities and to construct the shortest path among them.

## 3. The proposed drone static trajectory determination

In the path planning algorithm proposed in this paper, a single GPS-enabled drone is employed to localize stationary terrestrial nodes which are distributed uniformly in a wireless sensor network. We assume that the drone flies at a fixed altitude ($h$) above the terrestrial region of interest on the proposed static path and broadcasts beacons at the pre-determined points. This drone is able to broadcast beacon with different levels of transmission power and is utilized in the range-based localization mission.

The line of site distance between the drone and the receiver sensor, $d$ in Fig. 1, is called slant distance and its projection onto the ground, $d'$ in Fig. 1, is ground distance. If the projection of the aerial anchor point onto the ground is defined as waypoint then, each terrestrial sensor is assumed to be covered by the drone at a given aerial anchor point if and only if it is in the coverage circle of the drone's corresponding waypoint.

In the RSS-based localization techniques, the RSS values from nearby reference nodes are collected to estimate the target node location [26] as below:

1. The covered sensors receive a beacon including the current coordinate, and the transmission power of the drone.
2. The received information is used by the covered sensors to estimate the distances to the waypoints via measuring the received signal strength (RSS).
3. Finally, sensor nodes localize themselves based on all their estimated distances found through all the received beacons by trilateration or multilateration.

Here, we want to plan a trajectory which is energy conservative while trying to reduce the RSS-based location estimation error.

Using RSS for distance estimation is appealing since no additional hardware is necessary and distance estimates can even be derived without any additional overhead (apart from the communication itself that is taking place anyway [27]). The disadvantage, however, is that when using RSS as a ranging technique, it is necessary to accept and deal with considerable errors.

There are some solutions which can relieve the RSS-based estimation error. More received beacons at each sensor should mitigate the effect of ranging errors [27]. Furthermore, Ramadurai et al. [30] shows experimentally that longer distances lead to greater estimation errors. Authors in [31] also reports that by increasing the elevation angle between the drone and the sensors on the ground; the probability of having LOS connection increases. Thus, short distances between the fixed altitude drone and the ground user results in better localization output. Therefore, in this paper, we consider the number of received beacons at each sensor and their quality (which is determined by the sensor-to-drone distance) as two criteria in aerial anchor path planning.

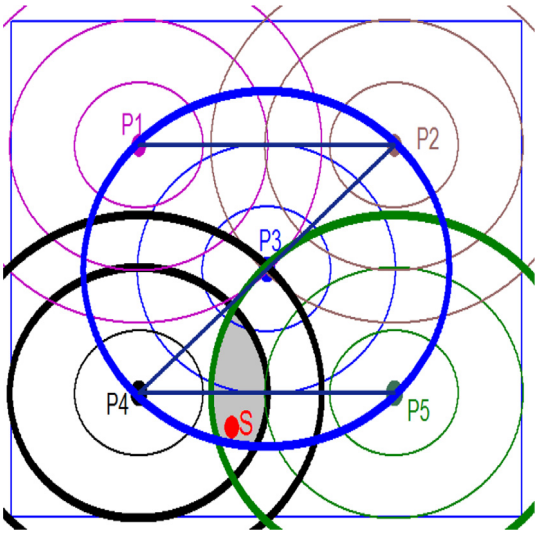



**Fig. 2.** Pre-localization step.

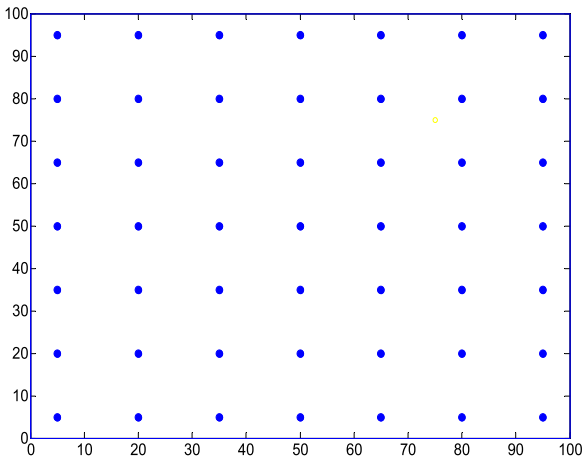



**Fig. 3.** Initial waypoints.

The main feature of the WSN is its highly dynamic nature. The sensors distribution may change due to energy depletion or unintentional displacement by wind or animals. Therefore, it is not realistic to consider that one-time localization of sensors can provide valid long-time location information. This makes energy consumption of the aerial anchor a determinant factor in trajectory planning. Consequently, a static path planning algorithm, called *WETAR*, is presented in this paper to guide the energy-constrained drone along the path efficiently.

Static *WETAR* determines the optimal path and the optimal transmission power of aerial anchor at each waypoint in two steps, namely the pre-localization and drone guiding steps. In the pre-localization step, a number of pre-defined positions which are located in the surveillance region are visited by the aerial anchor to gain information about sensors distribution. This information will be used during the drone guidance step in order to provide the maximum numbers of good quality beacons to the sensors with known estimative regions.

### 3.1. Pre-localization

A less energy-intensive pre-localization process is used where the overlap of circles indicate the sensors' estimative regions ($ER$). As shown in Fig. 2, the blue square represents the monitoring area. As

this figure shows, in the pre-localization step, the drone's projection traverses the surveillance region similar to Zcurve [13] while broadcasting beacons with different levels of transmission powers at waypoints $p_1$, $p_2$, $p_3$, $p_4$ and $p_5$. The dark blue line is the projected path followed by the drone and the circular regions indicate the communication ranges of the drone at these waypoints.

Upon receiving the beacon message, the sensor node understands that it is within the region of the circle centered at the coordinates of waypoint with a radius of $r$, where $r$ is the projected communication radius of the drone on the ground determined based on its transmission power. Based on the received beacon messages, a static sensor may reduce its estimative region. Upon reception of each new beacon, it does so by calculating the intersection of the estimative region of itself and the broadcast region of the new beacon. The shaded region in Fig. 2 shows the estimative region ($ER$) of sensor $S$ at the end of pre-localization.

Hereinafter, the drone would be aware of the $ERs$. The area size of the estimated region of each sensor is defined as the sensor location uncertainty.

Pre-localization step is used to determine estimative regions of sensors. They are utilized to assign weights to candidate beacon points (shown by dots in Fig. 3) in mobile anchor guiding step. The total path length traversed by the fixed height drone through its waypoints in this step is $(1+\frac{\sqrt{2}}{2})$L, where L is the side length of the terrestrial surveillance area.

### 3.2. Drone guidance algorithm

Network coverage and broadcasting good quality beacons are the two main issues which can greatly influence UAV-based localization. However, the energy-constrained drone which is considered in this paper, cannot guarantee full coverage. Instead, the objective of the weight-based optimization policy is to ensure localization of the maximum feasible number of sensors with the given energy budget. The definition of good quality beacon depends on the localization technique, i.e. range-based or range-free. In this paper, since range-based localization is utilized, closer waypoints to each sensor provide it with higher quality beacons.

We will use Integer Linear Programming (ILP) to find the most befitting waypoints and their corresponding transmission powers according to their assigned weights in the presence of sensors with given estimative regions. Dynamic behavior of the sensor network topology especially in outdoor environments is the main reason that makes the re-localization of sensors necessary. Therefore, the preassigned energy limitation is considered as a constraint in the selection of optimal waypoints. This novel aerial anchor path planning scheme is called Weighted Energy-aware Trajectory with Adaptive Radius (*WETAR*). *WETAR* path planner determines optimal path in two steps as follow:

(1) Weight Assignment

This step aims to assign weights to the waypoints which are shown in Fig. 3. Let $ER_i$ be the estimative region of the $i_{th}$ sensor $S_i$ (which is acquired in the pre-localization phase) and $CR\ (b_j, r_k)$ in Eq. (1) be the projected communication circle of the drone at the $j$th waypoint with a projected communication radius of $r_k$. $OA_{i,j}(r_k)$ in Eq. (2) shows their overlap area.

$$CR(b_j, r_k) = \{x, y | \left(x - x_{bj}\right)^2 + \left(y - y_{bj}\right)^2 \leq {r_k}^2\} \tag{1}$$

$$OA_{i,j}(r_k) = ER_i \cap CR(b_j, r_k) \tag{2}$$

Then, the coverage probability of sensor $S_i$ by the drone whose projection on the ground is $b_j$ and its projected communication radius is $r_k$ will be $prob_{S_i}(b_j, r_k)$, and is calculated by Eq. (3).

$$prob_{S_i}(b_j, r_k) = \frac{Area\ of\ (OA_{i,j}(r_k))}{Area\ of\ (ER_i)} \tag{3}$$

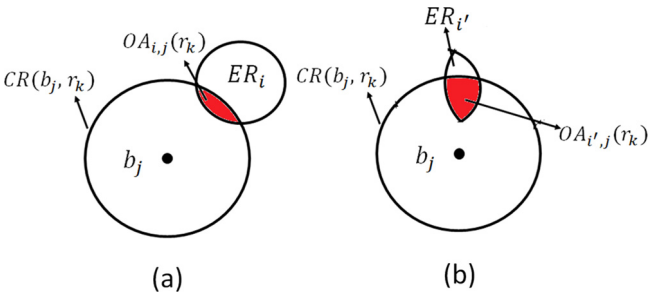



**Fig. 4.** Probability of coverage by the drone when it flies over the waypoint $b_j$ : (a) for sensor $S_i$, (b) for sensor $S_{i'}$.

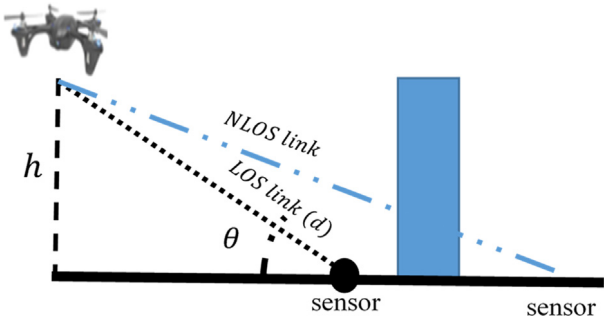



**Fig. 5.** LOS and NLOS links.

Fig. 4 explains Eq. (3) by an example. This figure shows if a sensor, $S_i$, is being located in overlap area of $ER_i$ and $CR\ (b_j, r_k)$ or sensor, $S_{i'}$, is being located in overlap area of $ER_{i'}$ and $CR\ (b_j, r_k)$ (which are shown by red color), it will receive the broadcasted beacon otherwise it will not be covered by the waypoint at $b_j$. Due to bigger $OA_{i',j}\left(r_k\right)$ in Fig. 4(b) than $OA_{i,j}(r_k)$ in Fig. 4(a), it is more probable that the broadcasted beacon form $b_j$ is received by sensor $S_{i'}$ than $S_i$.

To provide good quality beacons, the candidate waypoints are also weighted based on their distances from the covered sensors.

As shown in Fig. 5, the path loss between an aerial anchor and a terrestrial sensor can be determined according to the slant distance between the drone and its covered terrestrial sensor. Furthermore, the propagation environment (e.g. rural or urban) is a determinant factor in air to ground channel modeling since it influences the path loss. Accordingly, the drone to terrestrial sensor link can be either Line of Sight (LoS) or Non Line of Sight (NLoS). The environmental constraints such as the location of obstacles, their density and their height are unknown. As a result, most studies, including [31], assume a probabilistic LoS model. Eq. (4) shows the common probabilistic LoS model which is given in [31] and is derived using the statistical parameters provided by ITU-AR.

$$P_{LOS} = \frac{1}{1 + Cexp(-B\left[\theta - C\right])} \tag{4}$$

where C and B are constant values that depend on the environment and $\theta = \sin^{-1}\frac{h}{d}$ is the elevation angle in degree [29]. Therefore, the probability of having LOS link between the drone with fixed altitude ($h$) and terrestrial nodes increases in closer distances (d) due to the bigger elevation angles ($\theta$). Furthermore, closer waypoints leads to lower error in RSS-based distance estimation. Here, we discuss how the distance between the receiver sensor and the waypoint can affect the quality of the beacon. The basic model which is used in literature [23] for mapping RSS measurements to distance is defined in Eq. (5).

$$p_{RSS} = p_t - p_l\left(d_0\right) - 10\gamma \log\left(\frac{d}{d_0}\right) + X_{\sigma_{RSS}} \tag{5}$$

where, $p_{RSS}$ is the received signal power, $p_t$ is the transmit power of the sender, $p_l\left(d_0\right)$ is the power loss at reference distance $d_0$, $d$ is the

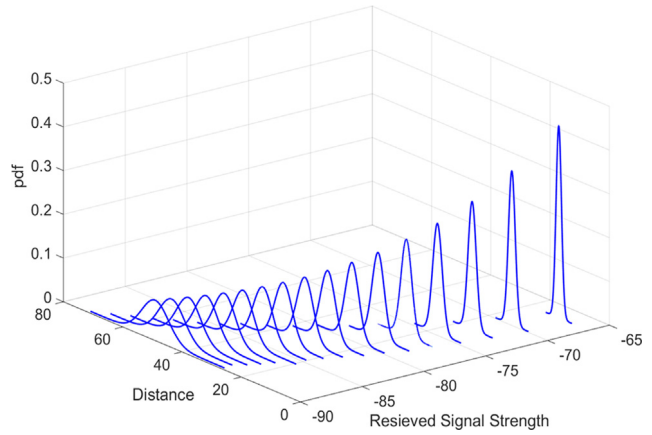



**Fig. 6.** Probability density functions of the estimated distances for various (mean) RSSI values.

real distance from the sender to the receiver, $\gamma$ is path loss exponent and $X_{\sigma_{RSS}}$ is a Gaussian random variable represents the noise in the measured RSS. An expression for the distance $d$ as a function of $p_{RSS}$ can be obtained by solving (5) for d, yields (6).

$$d = d_0 \times 10^{\frac{X_{\sigma_{RSS}} - p_l(d_0)}{10\gamma}} \times 10^{\frac{p_t - p_{RSS}}{10\gamma}} = const \times 10^{\frac{p_t - p_{RSS}}{10\gamma}} \tag{6}$$

The first-order Taylor series expansion of $d\left(p_{RSS}\right)$ is expressed in Eq. (7).

$$d\left(p_{RSS}\right) = d\left(\mu\right) + d'\left(\mu\right) \times \left(p_{RSS} - \mu\right) + remainder \tag{7}$$

Assume that $p_{RSS}$ satisfies the standard central limit theorem(CLT), then the $p_{RSS}$ is a random variable with mean value $\mu$. So, it follows that remainder is 0 in probability as well. By dropping the remainder Eq. (8) is obtained:

$$d\left(p_{RSS}\right) = d\left(\mu\right) + d'\left(\mu\right) \times \left(p_{RSS} - \mu\right) \tag{8}$$

The variance of $d\left(p_{RSS}\right)$ is calculated by the following Equation:

$$var\left(d\left(p_{RSS}\right)\right) \cong d'\left(\mu\right)^2 .var\left(p_{RSS}\right) \Rightarrow \sigma_d = d'\left(\mu\right).\sigma_{RSS} \tag{9}$$

Therefore, the standard deviation of estimated distance, $\sigma_d$, is directly proportional to the real value of distance as shown in Eq. (10).

$$\sigma_d = const.d.\sigma_{RSS} \tag{10}$$

Fig. 6 shows the probability distribution of estimated distance at each signal strength. It maps RSSI values to estimated distances for signal strengths ranging from −67 dBm to −89 dBm. The mean and standard deviation of estimated distances for each signal strength are calculated. As it is shown, distance estimations from farther beacons have higher standard deviations.

Hence, the weight of the waypoint $b_j$ and its distance from the covered sensor $S_i$ are reversely proportional. This concept is discussed in Fig. 7. In this figure solid elliptical region indicates estimative region of the sensor $S_i$, circular region shows communication area of the waypoint $b_j$ with communication radius $r_k$, and the blue-colored region, $OA_i$, shows the overlap of these two regions. The covered sensor $S_i$ may be located on any of the arcs in $OA_i$. The probability of being on $arc_m$ for sensor $S_i$ which is located in $OA_i$ is calculated by Eq. (11).

$$probability_{S_i}\left(arc_m\right) = \frac{length\ of\ arc_m}{\sum_{l=1}^{N_{ARC}} length\ of\ arc_l} \tag{11}$$

where, $N_{ARC}$ shows the total number of arcs in $OA_i$. The quality of beacon broadcasted at $b_j$ and received by sensor $S_i$ which is located on $arc_m$ is $\frac{1}{d_{arc_m - b_j}}$ where $d_{arc_m - b_j}$ shows the distance of $arc_m$ from the waypoint $b_j$. Therefore, the weight of the initial waypoint $b_j$ with communication radius $r_k$ as a result of sensor $S_i$ coverage is defined in Eq. (12).

$$W_{S_i, b_{j,k}} = prob_{S_i}(b_j, r_k)$$

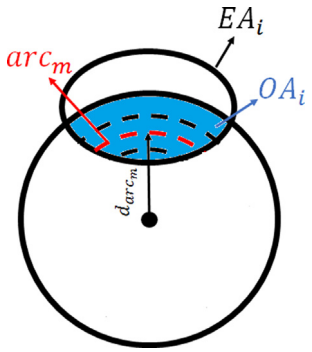



**Fig. 7.** Weight definition of a waypoint due to its covered sensor $S_i$.

$$\times \left[\sum_{m=1}^{N_{ARC}} \left(probability_{S_i}\left(arc_m\right) \times \left(\frac{1}{d_{arc_m - b_j}}\right)\right)\right] \tag{12}$$

All the sensors which are located in the communication range of the waypoint $b_j$ when its communication radius is equal to $r_k$ results in total weight of $W_{b_j, r_k}$ in Eq. (13) for $b_j$, $r_k$.

$$W_{b_j, r_k} = \sum_{\substack{i=1 \\ Area(ER_i \cap CR(b_j, r_k)) \neq 0}}^{N} W_{S_i, b_{j,k}} \tag{13}$$

(2) Trajectory Planning

In this section, the initial waypoints in Fig. 2 with various transmission powers or radius, $b_j, r_k$, are considered as the vertices of a weighted fully-connected graph $G$ where the weight of each edge is defined as the summation of assigned weights to its two vertices as follow:

$$EW_{i,j}\left(l,k\right) = W_{b_i, r_l} + W_{b_j, r_k} \tag{14}$$

It is not possible for an energy-constraint drone to fully cover the terrestrial monitoring region and broadcast beacons at all the initial waypoints. Moreover, receiving beacons from long distances results in inaccurate localization due to higher distance estimation error in RSS-based technique and also lower probability of LOS in UAV-based communication. Therefore, *WETAR* planner aims at the shortest trajectory construction for the done over the optimal waypoints while selecting their corresponding optimal communication radius.

If $E_{ij}(l,k)$ shows the edge of the graph $G$ between the vertices $b_{i,r_l}$ and $b_{j,r_k}$ and $EW_{i,j}\left(l,k\right)$ is its assigned weight, the drone path optimization problem aims at maximizing the total localization success, i.e. $TLS$ in (15), by sticking to the constraints presented in Eq. (16).

$$Max \quad TLS = \sum_{i=1}^{M-1} \left(EW_{i,j}\left(l,k\right) \times E_{ij}(l,k)\right) \tag{15}$$

Subject to:

$$\begin{aligned}
&\sum_{\substack{i=1 \\ i<j}}^{M-1} \sum_{l=1}^{L} \sum_{k=1}^{L} (E_{ij}(l,k)) \leq 1 \qquad \forall j = 2. \ldots .M \\
&\sum_{\substack{j=2 \\ i<j}}^{M} \sum_{l=1}^{L} \sum_{k=1}^{L} (E_{ij}(l,k)) \leq 1 \qquad \forall j = 2. \ldots .M-1 \\
&\sum_{i=1}^{M-1} \sum_{j=i+1}^{M} \sum_{l=1}^{L} \sum_{k=1}^{L} \left(\left[\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2\right]^{\frac{1}{2}} \right. \\
&\qquad \left. \times e_d + e_{BR}(l)\right) \times E_{ij}\left(l,k\right) \\
&\qquad \leq Energy_{MA} \\
&E_{ij}(l,k) \in \{0,1\}
\end{aligned} \tag{16}$$

**Table 1**
List of notations.

| Symbol | Description |
|---|---|
| $ER_i$ | Estimative region of the $i_{th}$ sensor $S_i$ |
| $r_k$ | Projected communication radius of the drone |
| $b_j$ | $j$th waypoint $(x_{bj}, y_{bj})$ |
| $CR\ (b_j,\ r_k)$ | Communication circle of the drone at the $j$th waypoint with a radius of $r_k$ |
| $OA_{i,j}(r_k)$ | Overlap area of $ER_i$ and $CR\ (b_j,\ r_k)$ |
| $prob_{S_i}(b_j,\ r_k)$ | Coverage probability of sensor $S_i$ by the drone at $b_j$ with communication radius of $r_k$ |
| $P_{LOS}$ | Likelihood of LOS link |
| $p_{RSS}$ | Received signal power |
| $p_t$ | Transmit power |
| $p_l\ (d_0)$ | Power loss at reference distance $d_0$ |
| $\gamma$ | Path loss exponent |
| $probability_{S_i}\ (arc_m)$ | Probability of being on $arc_m$ for sensor $S_i$ which is located in $OA_i$ |
| $N_{ARC}$ | Total number of arcs in $OA_i$ |
| $d_{arc_m - b_j}$ | Distance of $arc_m$ from the waypoint $b_j$ |
| $\frac{1}{d_{arc_m - b_j}}$ | The quality of beacon broadcasted at $b_j$ and received by sensor $S_i$ which is located on $arc_m$ |
| $W_{S_i,b_{j,k}}$ | The weight of the initial waypoint $b_j$ with communication radius $r_k$ as a result of sensor $S_i$ coverage |
| $W_{b_j,r_k}$ | The weight of the initial waypoint $b_j$ with communication radius $r_k$ as a result of all covered sensors |
| $E_{ij}(l,k)$ | Edge of the graph $G$ between the vertices $b_{i,l}$ and $b_{j,k}$ |
| $EW_{i,j}\ (l,k)$ | The assigned weight to edge of $E_{ij}(l,k)$ |
| $TLS$ | Total localization success |
| $M$ | Total number of initial waypoints |
| $L$ | Total number of transmission power levels |
| $e_d$ | Energy consumption of drone when it flies one meter over a fixed height plane |
| $e_{BR}(l)$ | Energy consumption of the drone when it broadcasts a beacon packet with $l$th transmission power level |
| $Energy_{MA}$ | Available energy of the drone in localization process |

where $M$ is the total number of initial waypoints and $L$ is the total number of transmission power levels, $e_d$ is the drone energy which is consumed when it flies one meter over a fixed height plane, $e_{BR}(l)$ is the energy consumption of the drone when it broadcasts a beacon packet with $l$th transmission power level, $Energy_{MA}$ is the available energy of the drone in localization process, and terrestrial location of $k$th waypoint is $(x_k, y_k)$. According to the two first constraints the drone is allowed to broadcast one beacon at most at each initial waypoint. The connectivity of the optimal resultant path is ensured by the third constraint. The last constraint limits the total energy consumption of the drone in the localization task to the value of $Energy_{MA}$. Table 1 summarizes our notations.

The linear cost function, $TLS$, linear equality and linear inequality constraints which are described in Eq. (16) makes ILP a feasible solution for *WETAR* optimization problem.

An integer linear program is a linear program that is further constrained by the integer outputs. The solution begins by solving an initial relaxed (non integer) problem using linear programming solutions. This paper uses the branch and bound method [32] to choose non-integer solutions of the relaxed problem. Then creates two new sub problems in which the value of that variable is more tightly constrained. These sub problems are solved and the process is repeated until a solution that satisfies all of the integer constraints is found. Finally, *WETAR* consists of selected waypoints and their corresponding transmission radius is determined as the set of edges on the graph $G$ in Eq. (17).

$$\text{Path} = \left\{ E_{\text{ij}}(\text{l},\text{k}) | E_{\text{ij}}(\text{l},\text{k}) = 1 \right\} \tag{17}$$

Fig. 8(a) to (d) show examples of *WETAR*-based paths for various values of mobile anchor energy which is a drone in this paper. Initial waypoints are indicated by small red circles, while the small red dots are shown selected waypoints. The localized and unlocalized sensors are shown by red and green stars, respectively and the blue line is the *WETAR* trajectory. The localized and unlocalized sensors are shown by red and green stars, respectively and the blue line is the *WETAR* trajectory. As these figures show, *WETAR* performs guiding task efficiently. It takes energy limitation into account and constructs the path among denser regions while it avoids empty regions.

## 4. Performance study

In this section, we evaluate the performance of *WETAR* through simulation. We conduct extensive simulation in an ideal environment without any obstacle. We also implement SCAN [19], Zcurve [13] and LMAT [20] with Matlab programming language to compare *WETAR* performance with these existing static paths considering different evaluation criteria.

### 4.1. Simulation setup

We assume a two-dimensional WSN including static sensors which are deployed randomly and a drone with a limited source of energy which is flying over the terrestrial area and localize ground objects using range-based approach.

Range-based location determination is done based on RSS values. The other parameters used in these simulations are listed in Table 2. WETAR trajectory planner can be applied to any shape for the deployment area, though without loss of generality, we assume a square-like monitoring region. Here, path loss is calculated according to Eq. (18) which is suggested in [33].

$$P_L = 33 \log \frac{d}{8} + 58.5 \tag{18}$$

We also assume that each sensor is equipped with CC2420 radio [34, 35]. Radio hardware of CC2420 offers a register to specify the power level during runtime.

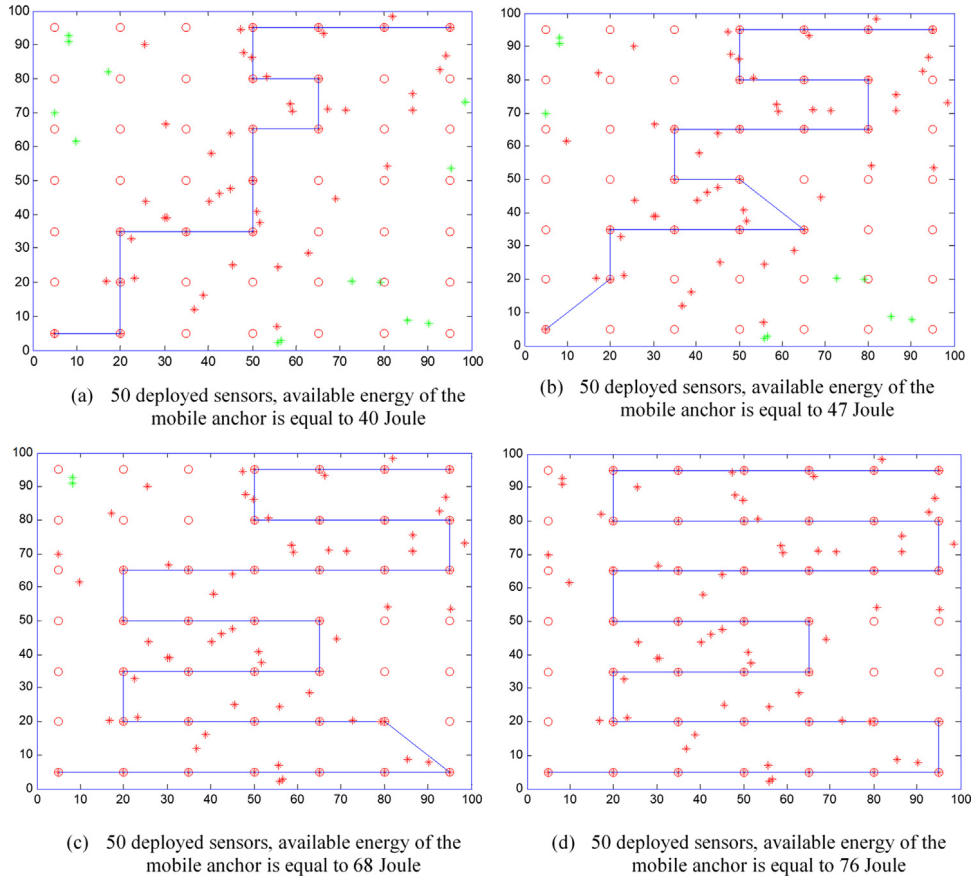
### 4.2. Simulation results

In the simulation study, localization coverage, localization precision and localization time are considered as means of comparison between *WETAR* and the existing static path planners.

#### 4.2.1. Localization coverage

We define the number of unknown terrestrial sensors that receive at least M non-collinear beacons from the drone while it flies over the determined trajectory as the localization coverage. M is a user-defined parameter which cannot be smaller than three. Larger values for M improves the location estimation error. Localization coverage of the various static path planners are compared for different values of energy constraints for the drone in Fig. 9. In this figure the communication radius $r_c$ of the drone in SCAN, LMAT and Zcurve trajectories is equal to 12.59 m. As this figure shows, the common behavior of all trajectories is the localization coverage increment as a result of increase in energy budget of the drone. However, regardless of the energy value of the drone, *WETAR* can localize more sensors than the other planners. The required energy of the drone to convince full localization coverage requisite in *WETAR* is less than the other path planning schemes. It is also concluded from this figure that the privilege of *WETAR* path in terms of localization coverage is more remarkable for lower values of energy.

Localization coverage is also studied under higher network density in Fig. 10. According to Figs. 9 and 10, the value of localization coverage improvement due to *WETAR* in comparison with the best existing trajectory increases from 77.24% to 88.54% as the sensor density increases. The major reason for superiority of proposed approach is that it constructs the drone trajectory over the optimal waypoints which are selected based on their contribution in localizing sensors. However, in SCAN and LMAT scenario if the drone broadcasts beacons with communication radius of 35.86 m they will outperform the *WETAR* in terms of localization coverage as Fig. 11 shows but they result in high value of location errors (more than 15 m) in comparison with *WETAR* with lower values of location error (less than 4 m).



(a) 50 deployed sensors, available energy of the mobile anchor is equal to 40 Joule

(b) 50 deployed sensors, available energy of the mobile anchor is equal to 47 Joule

(c) 50 deployed sensors, available energy of the mobile anchor is equal to 68 Joule

(d) 50 deployed sensors, available energy of the mobile anchor is equal to 76 Joule

**Fig. 8.** WETAR-based trajectory with different values of available energy for the drone as an aerial mobile anchor.

#### 4.2.2. Localization precision

The mean location error of the localized sensors is investigated as a representative of the localization precision achieved by the drone following various paths.

Fig. 12 plots the mean location error achieved in drone-based localization with different values of available energy following different static trajectory planners. Localization precision is improved by increasing the available energy of the drone as a result of higher number of provided beacons at each sensor. However, this is not true for some cases especially for Zcurve trajectory. Because it is more likely that sensors receive beacons from far distances which result in higher RSS noise.

Fig. 12 also performs simulations in networks with different densities to analyze the impact of sensors density on their resultant mean location error. The privilege of *WETAR* in terms of estimated location precision in comparison with the other planners is remarkable, especially in the case of high network density. As it is shown, guiding the drone along the *WETAR* path reduces the location error 67.95% to 99.25% and 81.3% to 99.25% for 50 and 150 terrestrial sensors, respectively. The reason behind this reduction is that in the existing static trajectories, path determination of the drone is done regardless of the sensors distribution while the *WETAR* trajectory and transmission radius at each selected waypoint are selected optimally with respect to their well-defined weights in Section 3.2.

Fig. 13 plots the mean location error in terms of drone energy under communication radius of 35.86 m for SCAN, LMAT and Zcurve path planning. It is observed that localized sensors experience higher location error in SCAN, LMAT and Zcurve when communication radius of the drone increases. This weak performance is the result of received beacons from far distances that leads to larger RSS noise.

**Table 2**
Simulation parameters.

| Parameter | Value |
|---|---|
| Terrestrial surveillance area | 100 m × 100 m |
| Sensor distribution | Uniform |
| Density of sensors | 0.005, 0.01, 0.015 (sensors/m$^2$) |
| Transmission power | $-25$ dBm $\leq P_t \leq -5$ dBm |
| Beacon packet size | 20 Byte |
| Path loss exponent ($\gamma$) | 3.3 |
| Data rate | 250 kbps |
| Energy to move drone over one meter | 0.1 J |
| Number of simulation runs | 500 |

#### 4.2.3. Localization time

The time needed for stopping and broadcasting beacons is calculated by Eq. (19) and traveling time of the drone in Eq. (20) determine the localization time.

$$T_B = N_B \times t_b \tag{19}$$

$$T_t = \frac{L_t}{v} \tag{20}$$

where $N_B$ is the number of beacon points and $t_b$ is the time needed for broadcasting a beacon. $T_t$ is the traveling time of the drone with constant speed of $v$ and traveling distance of $L_t$. Therefore, the localization time $T_L$ can be calculated as follow:

$$T_L = \left(N_B \times t_b\right) + \frac{L_t}{v} \tag{21}$$

Table 3 summarizes the drone energy that is consumed, the localization time, the trajectory length and the achieved location error for a full-coverage localization scenario. As the values in Table 3 indicate, WETAR planner provides a more energy-efficient path for the drone

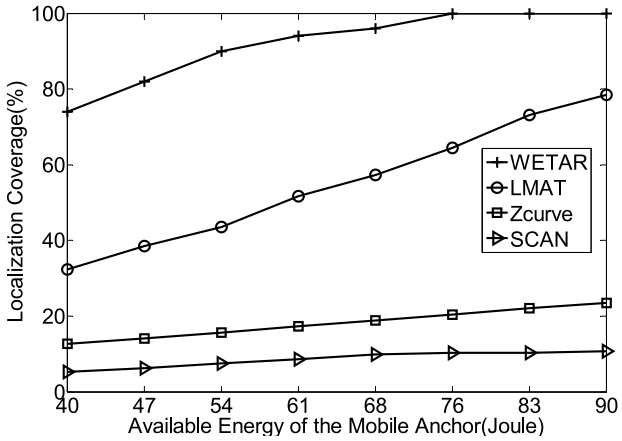



**Fig. 9.** Localization Coverage of 50 deployed sensors ($r_c$ is set to 12.59 m for SCAN, LMAT and Zcurve).

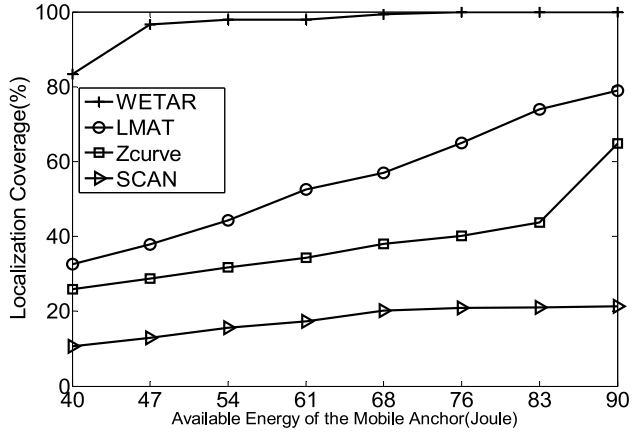



**Fig. 10.** Localization Coverage of 150 deployed sensors ($r_c$ is set to 12.59 m for SCAN, LMAT and Zcurve).

compared to the other static trajectories. The proposed method also spends less time on localization and traverses shorter paths and results in lower mean location error in comparison with other schemes.

#### 4.2.4. Helpless WayPoints rate

Another metric in range-based localization which can evaluate the drone trajectory efficiency is the rate off helpless waypoints which is described in Eq. (22). It indicates the number of waypoints that broadcasted beacons at their coordinates are not received by any terrestrial sensors. Therefore, they have no contribution in localization of WSNs.

$$HBR = 1 - \frac{\#\text{of useful beacon points}}{\text{total number of beacon points}} \tag{22}$$

As the simulation results in Fig. 14 shows, the *WETAR* trajectory eliminates helpless waypoints by assigning weight value of zero to them. Useless beacons in SCAN, LMAT and Zcurve schemes increase the energy consumption of the drone while they do not have any contribution to the progress of the localization task.

## 5. Conclusion

To our knowledge, this is the first study of UAV-based localization process that in which estimative regions of unknown terrestrial sensors are calculated in pre-localization phase. The estimative regions information are used to define weights for the pre-considered initial waypoints of the drone according to their localization coverage and the quality of beacons that they can provide for sensors to perform

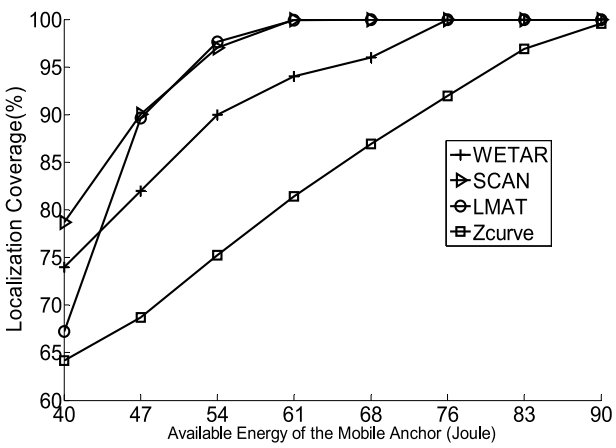



**Fig. 11.** Localization Coverage of 50 deployed sensors ($r_c$ is set to 35.86 m for SCAN, LMAT and Zcurve).

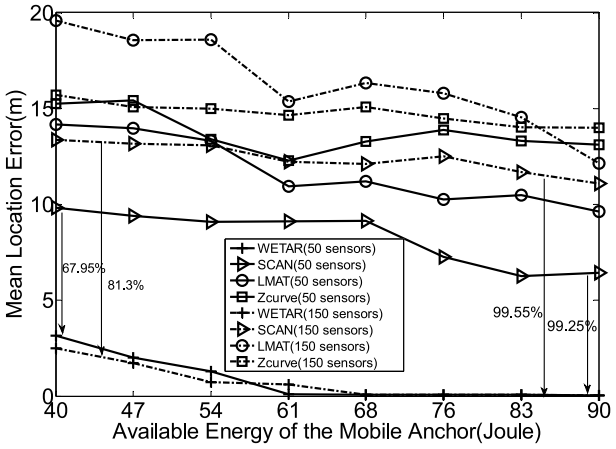



**Fig. 12.** Mean location error ($r_c$ is set to 12.59 m for SCAN, LMAT and Zcurve).

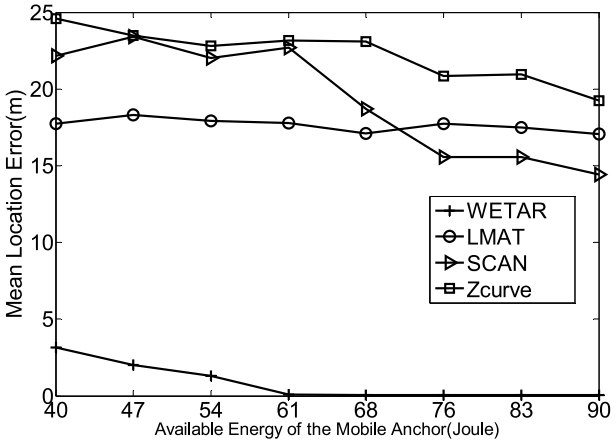



**Fig. 13.** Mean location error of 50 deployed sensors ($r_c$ is set to 35.86 m for SCAN, LMAT and Zcurve).

rang-based localization. Then the path planning problem for an energy-constraint drone is modeled as an optimization problem with a linear objective function and linear constraints. Finally, *WETAR* trajectory which includes the best waypoints and their corresponding transmission power is found using Integer Linear Programming (ILP) approach. Simulation results show that the *WETAR* trajectory planner can improves localization coverage, localization precision while reducing the

**Table 3**
Drone energy and localization time needed plus location error achieved in path planning mechanisms for range-based localization approach with full localization coverage ratio (with 100 sensors, v_drone = 2 m/s and the communication radii set to 12.59 m for SCAN, LMAT and Zcurve).

| Trajectory scheme | Energy (J) | Localization time (s) | Location error (m) | Trajectory length (m) |
|---|---|---|---|---|
| WETAR | 54 | 279 | 0.07 | 540 |
| LMAT | 155 | 777.66 | 12.35 | 1550 |
| SCAN | 220 | 1100 | 5.82 | 2200 |
| Zcurve | 232 | 1158 | 9.95 | 2320 |

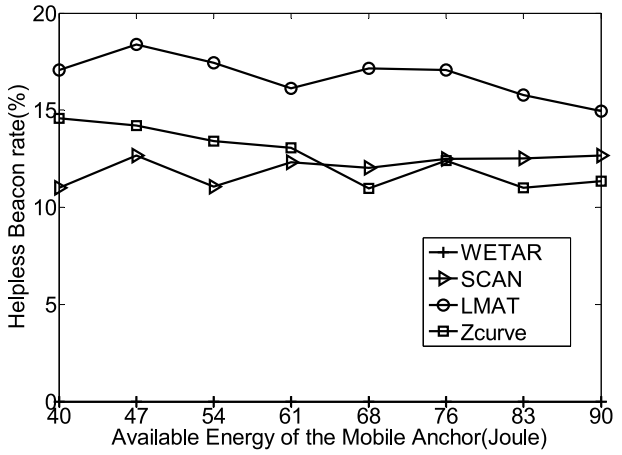



**Fig. 14.** Helpless beacons rate for 50 deployed sensors.

localization time in comparison with well-known static trajectories such as LMAT, Zcurve and SCAN. In this paper, we consider an ideal flying environment for the drone which is obstacle-free. We leave the analysis of UAV-based localization of terrestrial sensors in a non-ideal environment with the presence of obstacles as a future work.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Sahar Kouroshnezhad:** Conceptualization, Methodology, Software, Investigation, Visualization, Writing - original draft, Validation. **Ali Peiravi:** Conceptualization, Supervision, Validation. **Mohammad Sayad Haghighi:** Conceptualization, Supervision, Investigation, Writing - review & editing, Validation. **Alireza Jolfaei:** Project administration, Investigation, Resources, Validation.

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